

Radio Environment Maps Generation and Spectrum Sensing Testbed for Spectrum Sharing in 5G Networks

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Abstract — To deliver on the data rate and latency promises of 5G, more spectrum resources below 6 GHz are required. Therefore the regulators are now promoting spectrum sharing. Having access to realtime spatio-temporal spectrum usage information enables efficient decision making and monitoring. This knowledge can be represented in Radio Environment Maps constructed through geolocation aware spectrum measurements. In this paper, we give an overview of the spectrum sharing concept and its emergence in 5G standardisation. We also present our research on spectrum sharing including methods for efficient and accurate generation of Radio Environment Maps and a practical radio spectrum measurements testbed.

1 Introduction

On the road towards the next generation wireless communications (5G), we are aiming at massive number of subscriptions, ultra low latency communications and extremely high volume of data. This requires access to more spectrum below 6 GHz. Spectrum re-farming being increasingly difficult, spectrum sharing is believed to be a key technology that enables 5G.

Recently, regulators in Europe and the U.S. have released the spectrum sharing frameworks allowing the usage of the spectrum resources under a secondary license for the cellular networks. Two prominent spectrum sharing schemes are 1) Licensed Shared Access (LSA) and 2) Spectrum Access System (SAS). LSA is being developed by ETSI in Europe for the 2.3-2.4 GHz band [1]. SAS focuses on the 3.55-3.7 GHz band in US, and it is being developed by FCC [2, 3]. Both schemes open access to previously restricted mobile communications bands below 6 GHz. LSA proposed a two-tier sharing architecture between incumbents and licensees. Mobile Network Operators (MNOs) get access to the licensed spectrum when incumbents are absent at the time in the same geo-location area. There will be no co-channel interference between

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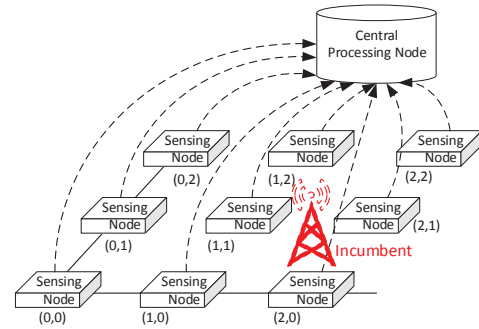


Figure 1: Architecture of the REM generation platform

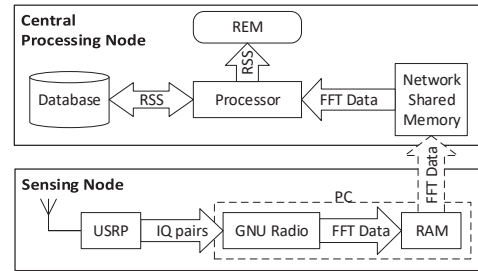


Figure 2: REM generation process

incumbents and licensees. In contrast, SAS proposed a three-tier sharing model. Incumbent users represent the highest tier in the framework and receive interference protection from Citizens Broadband Radio Service users. The Citizens Broadband Radio Service itself consists of two tiers - Priority Access and General Authorised Access (GAA) both authorised by SAS.

The FCC requires that information on incumbent services (military naval radar systems) to be undisclosed to MNOs while maintaining the protection of such military services from secondary emissions. The SAS architecture in its current form requires an independent dedicated third party entity named Environmental Sensing Capability (ESC) that detects the presence of incumbents within any part of

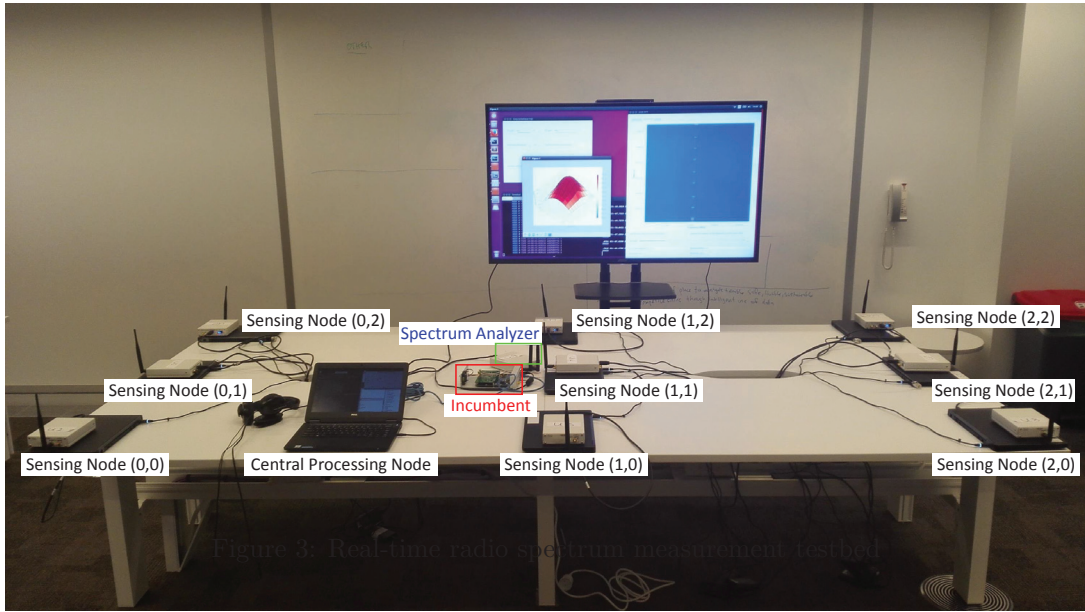


Figure 3: Real-time radio spectrum measurement testbed

3.55 - 3.7 GHz band. The incumbent being military radar systems, ESC is expected to comply stringent privacy requirements - hence the need for an independent entity.

Traditional frequency-time based sensing only provides local information, which is not sufficient in managing interference to and from a cellular network spread over a large area. Rather, spectrum sensing should be focused on finding both when and where the incumbents are active (sensing in time, frequency and space domains). Therefore, we propose a Radio Environment Maps (REM) generation architecture to sense the spectrum from different geo-locations, at the same frequency band and at the same time. Moreover, we setup a real-time REM generation testbed with software-defined radio and Universal Software Radio Peripherals (USRP) devices to measure the incumbent's received signal strength (RSS) in a laboratory environment.

2 Radio Environment Maps Generation

Consider a set of discrete N_s points in 2-D space covering a region where incumbents could be active. We allow k incumbents to actively transmit in a subset of k sparsely distributed discrete reference points. N_s number of sensing nodes will collect geolocation-tagged RSS measurements of the incumbent signal. Geolocation-tagged RSS measurements are then processed at a central location to determine at which reference points the incumbents are active and the interference level to the cellular networks.

In this paper, we assume that all the sensing nodes are on-grid and we have one node for each grid point, $N_s = N_p \times N_p$, where N_p is the grid width and length. In practical scenario, the sensing nodes can be off-grid. For the off-grid deploy-

ment, we have proposed a more complex theoretical approach to construct REM [4]. We use the RSS measurements for the REM values. The process of REM generation is shown in Fig. 2. Each sensing node receives the signal in time samples. To obtain the RSS for the REM, we convert the time domain signals into frequency domain signals. However, this conversion will consume heavy computational resources if it is conducted on the central processing node. Therefore, we propose that each sensing node does a distributed pre-process that is to use Fast Fourier Transform (FFT) to convert the time domain samples into frequency domain data. After that, each sensing node will send the FFT data to the network shared memory to which all the nodes have access. The central processing node will then fetch the FFT data of all sensing nodes from the network shared memory. The refresh rate of the REM can be as high as the sampling rate of the sensing node divided by the FFT size N_{FFT} . However, in practical scenario or in visual demonstration, we decrease the refresh rate of the REM to every N_r groups of FFT data. The RSS value at each grid point is calculated as

$$RSS = \frac{1}{N_r} \sum_{j=1}^{N_r} \frac{1}{N_{FFT}} \sum_{i=1}^{N_{FFT}} Y(i) \quad (1)$$

where $Y(i)$ is the FFT data in the network shared memory from each sensing node.

3 Real-time Radio Spectrum Measurement Testbed

We set up the platform with 9 sensing nodes, an incumbent, and a central processing node. We placed the sensing nodes in a 3x3 grid, each having the coordinates as shown in Fig. 1. The transmitter was placed in the centre of this grid. This testbed is

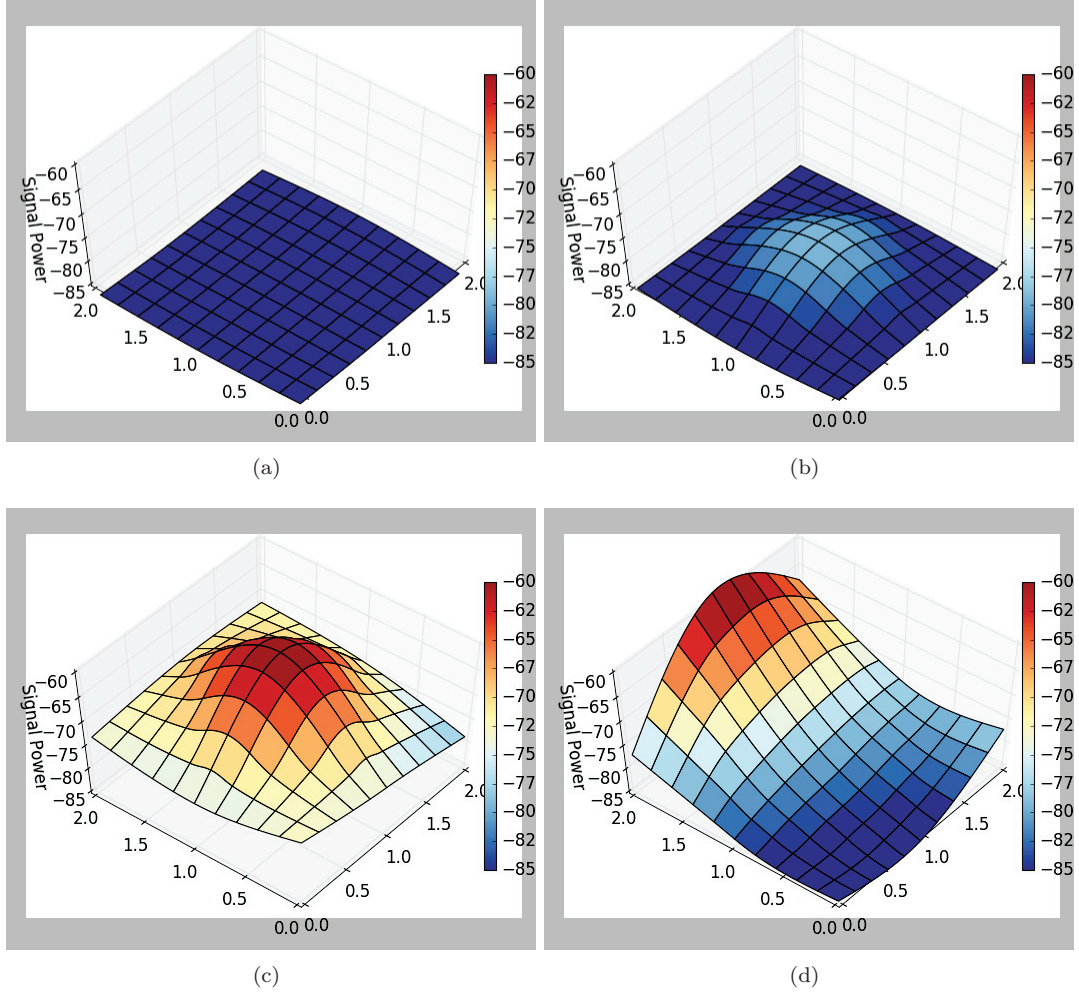


Figure 4: (a) REM when the incumbent is not transmitting. (b) REM when the incumbent is transmitting with 0dB gain. The location of the incumbent is close to the sensing node in the middle of the grid at (0,0). (c) REM when the incumbent is transmitting with 20dB gain. The location of the incumbent is close to the sensing node in the middle of the grid at (0,0). (d) REM when the incumbent is transmitting with 20dB gain. The location of the incumbent is close to the sensing node on the edge at (2,1).

shown in Fig. 3. The incumbent was a combination of a GNU radio installed laptop and a USRP2 as a transmitter. The sensing nodes were combinations of GNU radio installed laptops and USRP B200, tuned to the same band as receivers. We also set up a spectrum analyser next to the incumbent using USRP B200.

The data for all the grid points in one REM need to be the RSS from all sensing nodes at the same time instance. Therefore, all the nodes need to be synchronised with the same clock. We utilise Network Time Protocol (NTP) to adjust the clock of all the sensing nodes to be aligned with the central processing node. The accuracy of NTP varies based on the accuracy of the time source and the quality of the network connection used. Although

the synchronisation inaccuracy could vary between 5-100 ms [5], in our setup it was at most 20 ms. With NTP, we can guarantee that the data from all sensing nodes are synchronised.

Measurement results from sensor nodes were interpolated using splines. Although the interpolation in its current form does not utilise the correlation characteristics of fading, techniques such as Kriging interpolation can be utilised for this purpose. The number of sensors required to accurately construct a REM can be reduced by exploiting such characteristics.

4 Measurement Results

We tested our platform with two incumbent configurations. Firstly, we use a Binary Phase Shift Key-

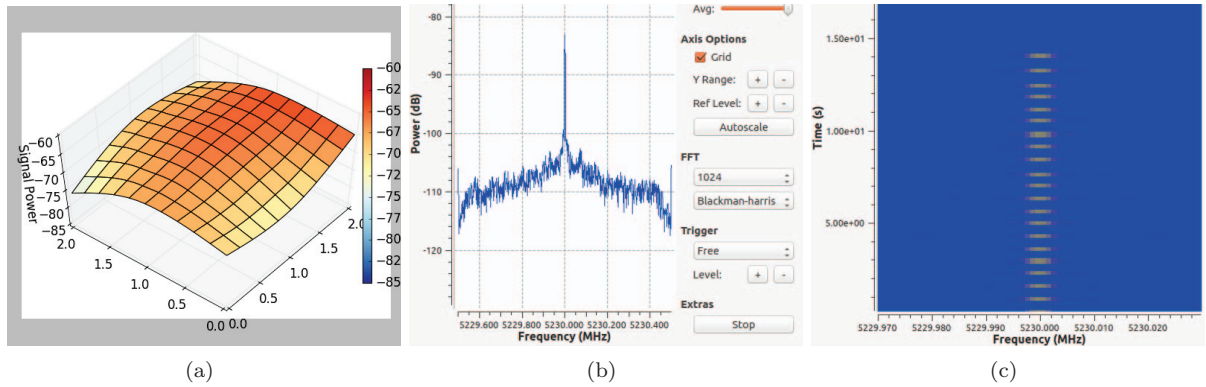


Figure 5: (a) REM when the measurement testbed capture the peak of the radar pulse and the radar is transmitting with 30dB gain. The location of the incumbent is close to the sensing node in the middle of the grid at (0,0). (b) The power spectrum of the radar signal. (c) The waterfall of the radar signal

ing (BPSK) Orthogonal Frequency Division Multiplexing (OFDM) 5 MHz bandwidth transmitter as the incumbent at the central frequency 5230 MHz. We set the incumbent to continuously transmitting the signal. This is a clear unlicensed band where no other transmitter is running. The sensing nodes are tuned to the same central frequency with 20 MHz bandwidth. Then the receiver is set to filter only the central 5 MHz signal, calculate the FFT data and send to the network shared memory. We show the measurement results of the OFDM incumbent signal in Fig. 4. We can see that when the incumbent is on, there is an obvious difference between when the incumbent is off which the measurement is only the noise floor. We can also see that our testbed can capture the changes of the location of the incumbent. Secondly, we use a Radar pulse transmitter as the incumbent. The interval between pulses is around 0.5 s. We show the measurement results and the signal analyser's result in Fig. 5. We took a snapshot of the REM when it shows the peak power of the radar pulse, the radar power spectrum and the waterfall. Our testbed can successfully capture the continuous transmitting incumbents, as well as the fast changing incumbents, such as radars.

The processing period is every 10 groups of FFT data, which is approximately 180 ms. Our testbed can run continuously without queuing the unprocessed data. Therefore, the processing delay is much less than 180 ms.

5 Conclusion

Spatio-temporal spectrum sensing through Radio Environment Maps is a key enabler of the 5G spectrum sharing systems. This paper presented methods to generate Radio Environment Maps with

geolocation aware measurements, and a sensing testbed constructed using software defined radios. The realtime radio spectrum measurement testbed is effective even in challenging applications such as detecting short radar pulses.

References

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